



Pancreatic diseases

Acute pancreatitis

- Acute pancreatitis accounts for 3% of all cases of abdominal pain admitted to hospital.
- It is a potentially serious condition with an overall mortality of 10%.
- About 80% of all cases are mild and have a favourable outcome.
- Approximately 98% of deaths from pancreatitis occur in the 20% of patients with severe disease and about one-third of these arise within the first week, usually from multi-organ failure.
- After this time, the majority of deaths result from sepsis, especially that complicating infected necrosis.

Pathophysiology of acute pancreatitis

- Acute pancreatitis is auto-digestion by protease enzyme as a result of premature activation of zymogen granules releasing which digest the pancreas and surrounding tissue.
- The normal pancreas has only a poorly developed capsule, and adjacent structures, including the common bile duct, duodenum, splenic vein and transverse colon, are commonly involved in the inflammatory process.
- The severity of acute pancreatitis is dependent upon the balance between activity of released proteolytic enzymes and anti-proteolytic factors like trypsin inhibitor protein .

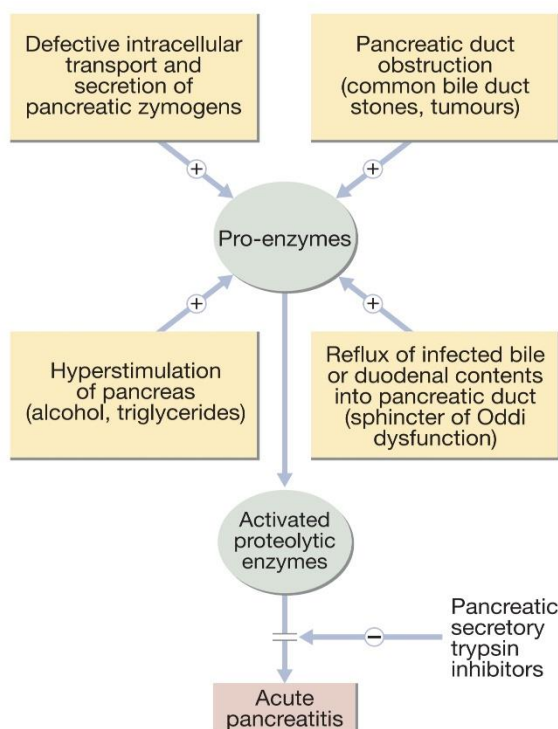


Fig. 21.64 Pathophysiology of acute pancreatitis.

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21.80 Causes of acute pancreatitis

Common (90% of cases)

- Gallstones
- Alcohol
- Idiopathic causes
- Post-ERCP

Rare

- Post-surgical (abdominal, cardiopulmonary bypass)
- Trauma
- Drugs (azathioprine/mercaptopurine, thiazide diuretics, sodium valproate)
- Metabolic (hypercalcaemia, hypertriglyceridaemia)
- Pancreas divisum (p. 842)
- Sphincter of Oddi dysfunction
- Infection (mumps, Coxsackie virus)
- Hereditary factors
- Renal failure
- Organ transplantation (kidney, liver)
- Severe hypothermia
- Petrochemical exposure

(ERCP = endoscopic retrograde cholangiopancreatography)



Causes of acute pancreatitis

Common (90% of cases)

1. Gallstones
2. Alcohol
3. Idiopathic
4. Post-ERCP

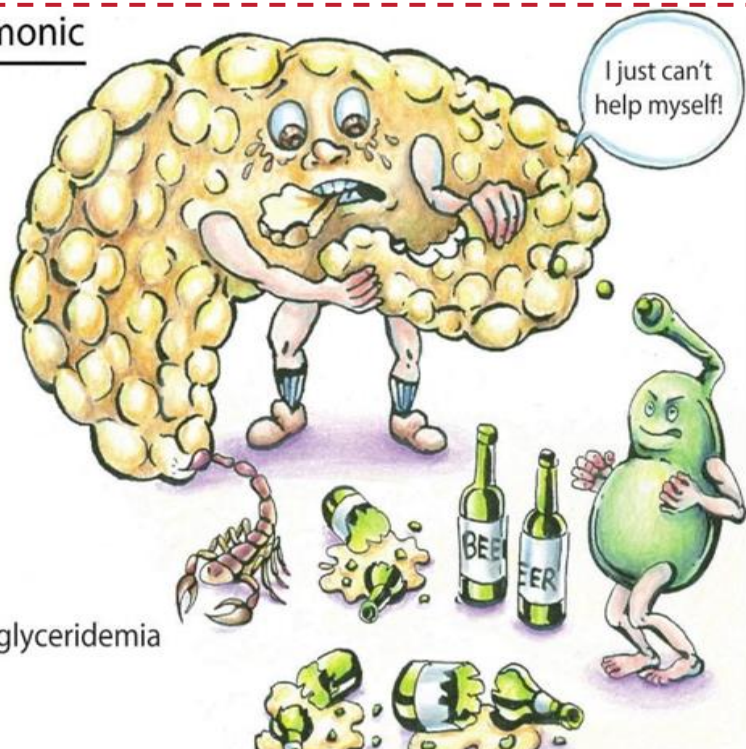
Less common causes :

- Post-surgical (abdominal, cardiopulmonary bypass)
- Trauma
- Drugs like: azathioprine, thiazide diuretics, L-asparaginase, and sodium valproate.
- Metabolic (hypercalcaemia, hypertriglyceridaemia)
- Pancreas divisum
- Sphincter of Oddi dysfunction
- Infection (mumps, Cocksackie virus)
- Hereditary
- Renal failure
- Organ transplantation (kidney, liver)
- Severe hypothermia
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I GET SMASHED Mnemonic

Pancreatitis

Idiopathic
Gall Stones
Ethanol (Alcohol)
Trauma
Steroids
Mumps / **M**alignancy
Autoimmune
Scorpion Stings
Hypercalcemia / **H**ypertriglyceridemia
ERCP
Drugs





Clinical features

- The typical presentation is with severe, constant upper abdominal pain, of increasing intensity over 15–60 minutes, which radiates to the back & decreased by bending forward.
- Nausea and vomiting are common.
- There is marked epigastric tenderness, but in the early stages (and in contrast to a perforated peptic ulcer), guarding and rebound tenderness are absent because the inflammation is principally retroperitoneal.
- Bowel sounds become quiet or absent as paralytic ileus develops.
- In severe cases the patient becomes hypoxic and develops hypovolaemic shock with oliguria.
- Discoloration of the flanks (Grey Turner's sign) or the periumbilical region (Cullen's sign) are features of severe pancreatitis with haemorrhage **“Haemorrhagic pancreatitis”**.

- The most important DDX of acute pancreatitis are:
- Perforated viscus.
- Acute cholecystitis.
- Inferior myocardial infarction



21.81 Complications of acute pancreatitis	
Complication	Cause
Systemic	
Systemic inflammatory response syndrome (SIRS)	Increased vascular permeability from cytokine, platelet-aggregating factor and kinin release
Hypoxia	Acute respiratory distress syndrome (ARDS) due to microthrombi in pulmonary vessels
Hyperglycaemia	Disruption of islets of Langerhans with altered insulin/glucagon release
Hypocalcaemia	Sequestration of calcium in fat necrosis, fall in ionised calcium
Reduced serum albumin concentration	Increased capillary permeability
Pancreatic	
Necrosis	Non-viable pancreatic tissue and peripancreatic tissue death; frequently infected
Abscess	Circumscribed collection of pus close to the pancreas and containing little or no pancreatic necrotic tissue
Pseudocyst	Disruption of pancreatic ducts
Pancreatic ascites or pleural effusion	Disruption of pancreatic ducts
Gastrointestinal	
Upper gastrointestinal bleeding	Gastric or duodenal erosions
Variceal haemorrhage	Splenic or portal vein thrombosis
Erosion into colon	Erosion by pancreatic pseudocyst
Duodenal obstruction	Compression by pancreatic mass
Obstructive jaundice	Compression of common bile duct

21.79 Features that predict severe pancreatitis	
Initial assessment	
• Clinical impression of severity • Body mass index >30 kg/m² • Pleural effusion on chest X-ray • APACHE II score >8 (see Box 10.50, p. 214)	
24 hours after admission	
• Clinical impression of severity • APACHE II score >8 • Glasgow score >3 (see Box 21.78) • Persisting organ failure, especially if multiple • CRP > 150 mg/L	
48 hours after admission	
• Clinical impression of severity • Glasgow score >3 • CRP > 150 mg/L • Persisting organ failure for 48 hours • Multiple or progressive organ failure	
(CRP = C-reactive protein)	

**COMPLICATIONS OF ACUTE PANCREATITIS****A. SYSTEMIC:**

1. Systemic inflammatory response syndrome (SIRS): Due to increased vascular permeability from multi-causes.
2. Acute respiratory distress syndrome (ARDS).
3. Hyperglycaemia.
4. Hypocalcaemia.
5. Low serum albumin.

B. PANCREATIC:

- Necrosis.
- Abscess.
- Pseudocyst.
- Pancreatic ascitis and/or pleural effusion.

C. GASTROINTESTINAL:

- Upper GIT bleeding due to gastric, duodenal erosions or varices.
- Duodenal obstruction due to pressure by pancreatic mass.
- Obstructive jaundice due to compression of CBD.
- Rarely erosion into colon.

PANCREATIC PSEUDOCYST

- **Pancreatic pseudocyst** is a localised peri-pancreatic collection of pancreatic juice and debris which usually develops in the lesser sac following inflammatory rupture of the pancreatic duct.
- The pseudocyst is initially contained within a poorly defined, wall of granulation tissue which matures over a 6-week period into a fibrous capsule.
- Small intrapancreatic cysts and pseudocysts are common features of both acute and chronic pancreatitis; they are usually asymptomatic and resolve as the pancreatitis recovers.
- Pseudocysts greater than 6 cm in diameter rarely disappear spontaneously.



Lec. 15

- **Large pseudocysts** cause constant abdominal pain, can produce a palpable abdominal mass and may compress or erode surrounding structures including blood vessels to form pseudoaneurysms.
- **Pancreatic ascites:** occurs when fluid leaks from a disrupted pancreatic duct into the peritoneal cavity.
- Leakage into the thoracic cavity can result in a pleural effusion or a pleuro-pancreatic fistula.

Diagnosis of acute pancreatitis

1. Typical clinical picture.
 2. Elevated enzyme levels. (amylase & lipase)
 3. Positive imaging. (US , CT scan) .
- **Elevation of S. amylase or lipase concentration:** S. amylase (normal range is 50-300 U/L) elevated early in the disease and soon return to normal due to renal excretion.
 - **Mild to moderate elevation also occurs in other acute abdominal conditions like**
 - A. Perforated DU.
 - B. Acute cholecystitis
 - C. Intestinal obstruction
 - D. Intestinal ischaemia
 - E. Ruptured ovarian cyst.
 - Urine amylase also may be measured.
 - A persistently elevated serum amylase concentration suggests pseudocyst formation !! .
 - If available, **serum lipase measurements are preferable to amylase**, as they have greater diagnostic accuracy for acute pancreatitis.
 - **Ultrasound scanning** can confirm the diagnosis, although in the earlier stages the gland may not be grossly swollen.
 - The ultrasound scan is also useful because it may show gallstones, biliary obstruction or pseudocyst formation.
 - **Plain X-rays** are taken to exclude other diagnoses such as perforation or obstruction and to identify pulmonary complications



- Contrast enhanced CT of abdomen (pancrease) after admission can be useful in assessing viability of the pancreas if persisting organ failure, sepsis or clinical deterioration is present, since these features may indicate that pancreatic necrosis has occurred.
- Necrotising pancreatitis is associated with decreased pancreatic enhancement on CT, following intravenous injection of contrast material.
- The presence of gas within necrotic material suggests infection and impending abscess formation, in which case percutaneous aspiration of material for bacterial culture should be carried out and appropriate antibiotics prescribed.
- Involvement of the colon, blood vessels and other adjacent structures by the inflammatory process is best identified by CT .
- Biochemical and haematological tests: like albumin, glucose, urea, ALT, lactic dehydrogenase, S. calcium, WBC count, C-reactive protein and Po2.
- (These are of prognostic value).
- Serial assessment of C-reactive protein (CRP) is a useful indicator of progress.
- It is worth noting that the serum amylase concentration has no prognostic value.

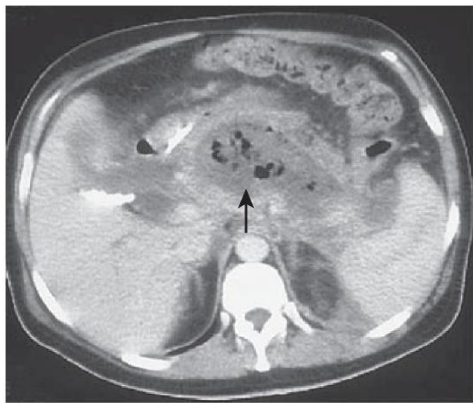


Fig. 21.66 Pancreatic necrosis. Lack of vascular enhancement of the pancreas during contrast-enhanced computed tomography indicates necrosis (arrow). The presence of gas suggests that infection has occurred.

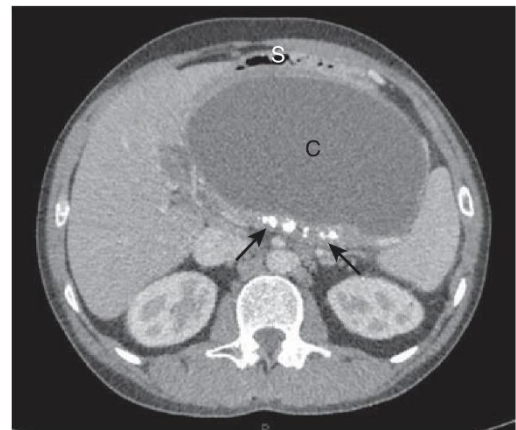


Fig. 21.65 Computed tomogram showing large pancreatic pseudocyst (C) compressing the stomach (S). The pancreas is atrophic and calcified (arrows).





MANAGEMENT OF ACUTE PANCREATITIS

▪ Principles of management :

1. Analgesia: Pethidine (but not morphine) may be needed for severe pain.
2. Correction of hypovolaemia using normal saline and/or colloids.
3. All severe cases should be managed in intensive care unit.
4. A central venous line or Swan-Ganz catheter and urinary catheter are used to monitor patients with shock.
5. Hypoxic patients need oxygen and patients who develop adult respiratory distress syndrome (ARDS) may require ventilatory support.
6. Hyperglycaemia is corrected using insulin.
7. It is not necessary to correct hypocalcaemia by intravenous calcium injection unless tetany occurs.
8. Nasogastric aspiration is only necessary if paralytic ileus is present.
9. Enteral feeding via a naso-enteral tube should be started at an early stage in patients with severe pancreatitis because
 - a. They are in a severely catabolic state and need nutritional support .
 - b. Enteral feeding decreases endotoxaemia and so may reduce systemic complications.
10. Prophylaxis of thromboembolism with low-dose subcutaneous heparin is also advisable.
11. Prophylactic, broad-spectrum intravenous antibiotics such as imipenem or cefuroxime may improve outcome in severe cases.

(Antibiotics is NOT routinely used !!)



SPECIFIC THERAPY

- Patients who present with cholangitis or jaundice in association with severe acute pancreatitis should undergo urgent ERCP to diagnose and treat choledocholithiasis.
- In less severe cases of gallstone pancreatitis, biliary imaging (using MRCP) can be carried out after the acute phase has resolved.
- If the liver function tests return to normal and ultrasound has not demonstrated a dilated biliary tree, laparoscopic cholecystectomy with an on-table cholangiogram is appropriate because any common bile duct stones have probably passed.
- When the operative cholangiogram detects residual common bile duct stones, these are removed by laparoscopic exploration of the duct or by post-operative ERCP.
- Cholecystectomy should be undertaken within 2 weeks following resolution of pancreatitis to prevent further potentially fatal attacks of pancreatitis.

Prognosis

- Adverse prognostic factors in acute pancreatitis (glasgow criteria):
 - Age > 55 years
 - P O₂ < 8 kPa (60 mmHg)
 - White blood cell count (WBC) > 15 × 10⁹/litre
 - Albumin < 32 g/l
 - Serum calcium < 2 mmol/l (8 mg/dl) (corrected)
 - S. Glucose > 10 mmol/l (180 mg/dl)
 - S. Urea > 16 mmol/l (45 mg/dl) (after rehydration)
 - Alanine aminotransferase (ALT) > 200 U/l
 - Lactate dehydrogenase (LDH) > 600 U/l



21.78 Glasgow criteria for prognosis in acute pancreatitis*

- Age > 55 years
- PO₂ < 8 kPa (60 mmHg)
- White blood cell count > 15 × 10⁹/L
- Albumin < 32 g/L (3.2 g/dL)
- Serum calcium < 2 mmol/L (8 mg/dL) (corrected)
- Glucose > 10 mmol/L (180 mg/dL)
- Urea > 16 mmol/L (45 mg/dL) (after rehydration)
- Alanine aminotransferase > 200 U/L
- Lactate dehydrogenase > 600 U/L

*Severity and prognosis worsen as the number of these factors increases. More than three implies severe disease.



Chronic pancreatitis

- Chronic pancreatitis is a chronic inflammatory disease characterised by fibrosis and destruction of exocrine pancreatic tissue.
- Diabetes mellitus occurs in advanced cases because the islets of Langerhans are involved.
- Pathologically chronic pancreatitis can be classified into 3 categories:
- (1) Chronic calcifying pancreatitis (almost always related to alcoholism),
- (2) Chronic obstructive pancreatitis the pancreatic duct is dilated, and the pancreas is normal in size, atrophic, or focally and/or globally enlarged.
- The most important D. D. of chronic pancreatitis is carcinoma of the pancreas

Causes of chronic pancreatitis

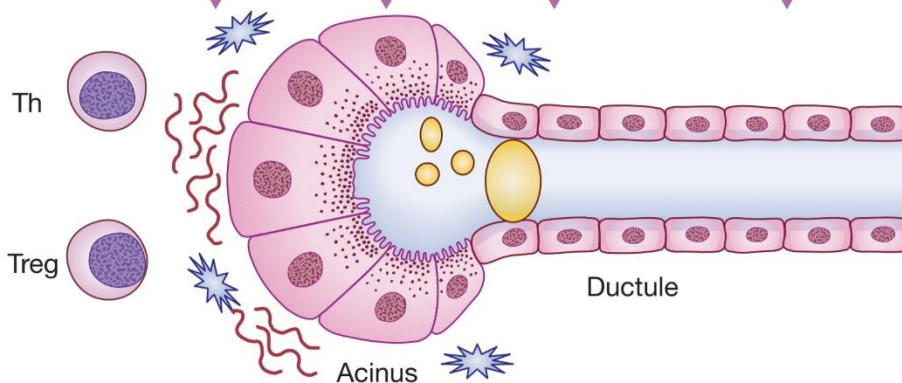
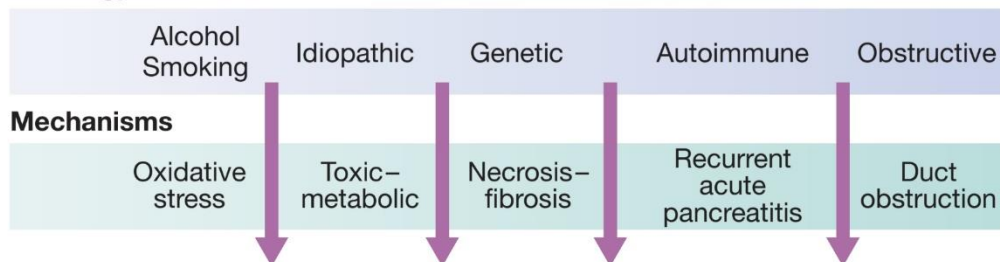
- The main causes of chronic pancreatitis include the following:
 1. Alcoholism: Changes appear to develop slowly and may develop after excessive alcohol consumption for 10 years or more.
Alcoholism is associated with chronic pancreatitis in 60-90% of patients.
 2. Idiopathic: Etiology is idiopathic in 10-40% of patients.
 3. Hereditary causes: Hereditary pancreatitis is an autosomal dominant disorder.
Symptoms usually appear in the patient's first decade of life, and eventually lead to both exocrine and endocrine pancreatic dysfunction
 4. Autoimmune conditions: Autoimmune pancreatitis is a rare condition that is often seen in patients with primary sclerosing cholangitis.
 5. Congenital causes: A congenital abnormality of fusion, pancreas divisum, can cause chronic pancreatitis
 6. Cystic fibrosis: This disease is associated with pancreatic atrophy and chronic pancreatitis
 7. Other conditions: Hyperlipidemia, hyperparathyroidism, and uremia can cause chronic pancreatitis.



21.82 Causes of chronic pancreatitis*	
Toxic–metabolic	
• Alcohol	• Hypercalcaemia
• Tobacco	• Chronic kidney disease
Idiopathic	
• Tropical	• Early-/late-onset types
Genetic	
• Hereditary pancreatitis (cationic trypsinogen mutation)	• <i>SPINK-1</i> mutation
	• Cystic fibrosis
Autoimmune	
• In isolation or as part of multi-organ problem	
Recurrent and severe acute pancreatitis	
• Recurrent acute pancreatitis	• Post-necrotic
Obstructive	
• Ductal adenocarcinoma	• Pancreas divisum
• Intraductal papillary mucinous neoplasia	• Sphincter of Oddi stenosis

*These can be memorised by the mnemonic 'TIGARO'. Gallstones do not cause chronic pancreatitis but may be observed as an incidental finding.

Aetiology



Clinical course



Fig. 21.67 Pathophysiology of chronic pancreatitis. Alcohol and other risk factors may trigger acute pancreatitis through multiple mechanisms. The first (or 'sentinel') episode of acute pancreatitis initiates an inflammatory response involving T-helper (Th) cells. Ongoing exposure to alcohol drives further inflammation but this is modified by regulatory T cells (Treg) with subsequent fibrosis, via activation of pancreatic stellate cells. A cycle of inflammation and fibrosis ensues, with development of chronic pancreatitis. Alcohol is the most relevant risk factor, as it is involved at multiple steps.



CLINICAL PRESENTATION

- Any age group can be affected, but chronic pancreatitis is unusual in children younger than 10 years.
 - Most patients are aged 40-50 years.
 - A slight male preponderance exists.
1. The most common presentation of chronic pancreatitis, is abdominal pain.
 - Pain can be episodic, lasting hours to days, or it can persist for months or even years.
 - The pain is characteristically steady in the epigastrium, and it frequently radiates to the back.
 2. Most patients lose weight during the course of the disease.
 - This weight loss may be related to malabsorption due to pancreatic exocrine deficiency or the fear of eating, because food intake exacerbates the pain.
 - Pain relieve may occur after alcohol drinking or leaning forwards.
 3. On occasion, the disease process can be painless, and patients may present with steatorrhea, weight loss, or diabetes mellitus.

Complications

1. Narcotic addiction is a common problem because of the intractable pain.
2. Weight loss and malabsorption are also common.
3. Pseudocyst and fistula formation.
4. Pseudoaneurysms of large arteries close to the pancreas.
5. Stenosis of the common bile duct.
6. Splenic and/or portal venous obstruction.
7. Diabetes can develop in 70-90% of patients with chronic calcific pancreatitis.
8. Pancreatic ascitis.
9. Peptic ulcer .



21.83 Complications of chronic pancreatitis

- Pseudocysts and pancreatic ascites, which occur in both acute and chronic pancreatitis
- Obstructive jaundice due to benign stricture of the common bile duct as it passes through the diseased pancreas
- Duodenal stenosis
- Portal or splenic vein thrombosis leading to segmental portal hypertension and gastric varices
- Peptic ulcer



INVESTIGATIONS

A. Tests to identify anatomy of the pancreas :

1. Plain radiographs show pancreatic calcification in 25-59% of patients.

This feature is pathognomonic for chronic pancreatitis.

NOTES: On anteroposterior radiographs, the spine may mask small punctate calcification; therefore, additional oblique or lateral imaging may be indicated.

2. Gastrointestinal (GI) tract barium testing still has a place in the management of chronic pancreatitis.

First, some patients with chronic pancreatitis present with atypical abdominal complaints, and initially, barium studies may be ordered. Second, complications from chronic pancreatitis may cause obvious changes in the GI tract.

3. Ultrasonography is the first modality to be used in patients presenting with upper abdominal pain, although the direct diagnosis of chronic pancreatitis is not always possible (the pancreas may appear normal even in the presence of advanced disease. In patients who are obese, excessive intraperitoneal gas may obscure the pancreas. Gas overlying the pancreas also can make visualization of the pancreas difficult). Sonography can help in determining the cause of chronic pancreatitis (eg, calculus disease) and in assessing the complications of the disease (eg, pseudocysts, ascites, splenic/portal venous obstruction

4. MRI, particularly MRCP, is a noninvasive technique.

MRI provides excellent images that may show the changes in the diseased pancreas and also the complications of chronic pancreatitis.

The use of secretin with MRCP can demonstrate pancreatic exocrine reserve. as well as dilated Santorini duct seen in pancreas divisum.

5. CT is excellent for imaging of the retroperitoneum, and it is useful in differentiating chronic pancreatitis from pancreatic carcinoma.



CT: shows an enlarged pancreas associated with punctate calcification

6. Retrograde Cholangiopancreatography (ERCP): Is the most sensitive imaging modality; it is used to show the ductal anatomy directly and when intervention (eg, stricture dilatation, stent placement) is contemplated
7. Angiography is reserved for patients with suspected angiographic complications resulting from chronic pancreatitis

B. Tests of pancreatic function for exocrine and endocrine functions like:

1. collection of pancreatic juice after secretin injection.
2. PABA (pancreolauryl) test .
3. Faecal chymotrypsin or elastase.
4. Blood glucose test.

i	21.84 Investigations in chronic pancreatitis
Tests to establish the diagnosis	
• Ultrasound • Computed tomography (may show atrophy, calcification or ductal dilatation) • Abdominal X-ray (may show calcification) • Magnetic resonance cholangiopancreatography • Endoscopic ultrasound	
Tests to define pancreatic function	
• Collection of pure pancreatic juice after secretin injection (gold standard but invasive and seldom used) • Pancreolauryl test (see Box 21.12, p. 777) • Faecal pancreatic elastase	
Tests to demonstrate anatomy prior to surgery	
• Magnetic resonance cholangiopancreatography	



Treatment of chronic pancreatitis

1. **Alcohol avoidance** is essential, although some patients try to relieve pain by drinking excessive alcohol which causes further pancreatic damages
2. **Surgical procedures** including partial pancreatic resection and pancreatico-jejunostomy.
3. **ERCP procedures:** including dilatation and stent insertion of main pancreatic duct and lithotripsy.
4. **Medical treatment:** Including enzyme replacement (exocrine function), treatment of diabetes mellitus by insulin therapy and the most important is pain relieving by giving different types of analgesics including narcotics which may lead to drug addiction. Pain may be relieved some times by giving oral pancreatic enzymes..
The medical treatment also includes vitamins and mineral replacement and treatment of malabsorption.
5. **Management of complications.**

i	21.85 Intervention in chronic pancreatitis
	Endoscopic therapy
	• Dilatation or stenting of pancreatic duct strictures • Removal of calculi (mechanical or shock-wave lithotripsy) • Drainage of pseudocysts
	Surgical methods
	• Partial pancreatic resection, preserving the duodenum • Pancreatico-jejunostomy



" لا تنسوا الدعاء لأهالينا في فلسطين "